ORIGINAL RESEARCH ARTICLE

ELSEVIER

Comparative Clinical Evaluation of the Effect on Carbohydrate and Lipid Metabolism of Two Norethisterone-Containing Hormonal Contraceptives: Mesigyna and TriNovum

G. Benagiano,* F.M. Primiero,* C. Bastianelli,* P. Bianchi,* and E. Medda†

The effect on carbohydrate and lipid metabolism of two hormonal contraceptive preparations containing norethisterone (commercially known as Mesigyna and TriNovum) was studied in a total of 60 women, before and after 6 months of treatment.

Carbohydrate metabolism was evaluated by means of a euglycemic glucose clamp test; lipid metabolism was monitored by measuring total cholesterol, HDL-cholesterol, LDL-cholesterol, VLDL-cholesterol, and triglycerides. The two groups were properly matched with the exception of pretreatment levels of cholesterol and LDL-cholesterol. At the end of treatment, no difference was found within or between groups in fasting glucose and insulin levels and in glucose rate of disappearance. A significant increase in total cholesterol, HDL-cholesterol, and VLDL-cholesterol was found in both groups at the end of the treatment period; in addition, TriNovum caused a significant increase also in triglycerides.

In conclusion, the safety of both preparations with regard to carbohydrate metabolism was confirmed using the most accurate method available; furthermore, changes in lipid metabolism were such as to have little clinical significance. CONTRACEPTION 1997;55:295–300 © *1997 Elsevier Science Inc. All rights reserved.*

KEY WORDS: carbohydrate metabolism, lipid metabolism, Mesigyna, TriNovum

*First Institute of Obstetrics and Gynecology, University "la Sapienza", Roma, Italy, and †Laboratory of Epidemiology and Biostatistics, Istituto Superiore di Sanitá, Roma, Italy

Name and address for correspondence: Dr. Francesco M. Primiero, 1° Istituto di Clinica Ostetrica e Ginecologina, Universitá la Sapienza, Policlinico Umberto 1°, 00161 Roma, Italy. Tel: 440.40.67

Submitted for publication October 25, 1996
Revised January 31, 1997
Accepted for publication February 17, 1997

Introduction

Since the early studies of Pincus and his group,[1] contraceptive effectiveness of estrogen–progestogen combinations (EP) has never been questioned. For this reason, the history of hormonal contraception in its combined form is the history of its adverse effects, drawbacks, and possible long-term consequences, as well as of the attempts, on the one hand to reduce their severity and frequency, and on the other to facilitate compliance. Compliance is an important problem with orally administered EP combinations, because it is relatively easy for a woman to forget to take her daily pill.

In order to improve contraceptive compliance, the idea was conceived more than 30 years ago[2] to administer long-acting preparations, with the double aim not only to improve continuation rate, but also to decrease dosage and therefore the frequency of adverse reactions. Although over the last three decades a number of monthly EP injectable preparations have been tested in clinical trials, fear of long-term adverse effects of long-acting estrogens has hindered the diffusion of this type of hormonal contraceptives.

This prompted the World Health Organization to promote a research program to develop a monthly combination utilizing a short-acting estrogen.[3] Studies on the pharmacokinetic profiles and pharmacodynamic effects of different estrogens[4] and various dosage combinations of a progestin and an estrogen led to the selection of estradiol valerate as the estrogen and of norethisterone enantate as the progestin. This new monthly injectable, known as Mesigyna, has been successfully tested in large multicenter clinical trials.[5–7]

Impairment of glucose tolerance and raised serum insulin concentrations have been often reported in women using early combined oral contraceptives.[8]

© 1997 Elsevier Science Inc. All rights reserved.
655 Avenue of the Americas, New York, NY 10010

ISSN 0010-7824/97/$17.00
PII S0010-7824(97)00033-4

Contraception
1997;55:295–300

The estrogenic component was initially believed to be responsible for these changes, until it was shown that estrogens bind to specific sites on the insulin receptor and prevent degradation of the insulin molecule,[9] thus promoting insulin biologic activity. For this reason, attention was focused on the progestogenic component of the pill, which interferes with insulin receptor binding and determines its decreased hepatic uptake.[10]

At the same time, several prospective epidemiological studies suggest a strong correlation between the risk of developing cardiovascular disease and significant changes in serum lipid profile.[11] In this context, the possible adverse effects induced by progestins, either alone or in combination with estrogens, have been extensively studied since the early 1980s,[12] the atherogenic potential being considered to be dependent upon the progestin's androgenic activity.

The present study was undertaken to monitor and compare the effects of the monthly injectable Mesigyna and those of the norethisterone-containing triphasic oral contraceptive, TriNovum, on the most sensitive test of carbohydrate disturbance and on specific parameters of lipid metabolism.

Materials and Methods

Hormonal Preparations

The following formulations were used:

1) Norethisterone enantate 50 mg + estradiol valerate 5 mg, administered intramuscularly every 30 days (Mesigyna);
2) Norethisterone 0.5, 0.75, 1 mg + ethinylestradiol 0.035 mg, given orally for 21 days (TriNovum).

Subjects

A total of 60 healthy female volunteers were screened and randomized to receive either Mesigyna or TriNovum. The study period included 8 cycles, 1 pretreatment, 6 treatments, and 1 post-treatment. To be included in the study, women had to be between 18 and 35 years of age; at least 12 weeks post-pregnancy, with no known or suspected diabetes, liver disease, or thromboembolic disorder; within 20% of their ideal body weight; and regularly menstruating. They were preliminarily properly informed of the requirements and aims of the study and accepted to participate.

Laboratory Procedures

The following tests were performed during the luteal phase of the pretreatment and during the last week of the 6th treatment cycle:

1) Euglycemic glucose clamp test: This consists of administering intravenously glucose and insulin until a steady state is reached, in which plasma glucose and insulin levels remain constant.[13] Results are expressed in glucose rate of disappearance (Rd mg/kg/min). The normal range in our laboratory is between 6 and 8, with a clinically significant insulin resistance observed with values <4. The variation coefficient (VC) of the test is the ratio between the mean glucose level at steady state and its standard error. Values >5 indicate an error in the performance of the test.
2) Lipid profile: the following measurements were made: total cholesterol, HDL-cholesterol, LDL-cholesterol, VLDL-cholesterol, and triglycerides (all expressed in milligrams/deciliter).[14–16] Samples were stored at −20°C and analyzed in two separate batches: pretreatment and the 6th month of treatment.

Statistical Methods

The validity of randomization between the groups was evaluated by homogeneity test. Within each study group, differences between the admission and the 6-month follow-up evaluation were assessed by paired *t*-test. Unpaired *t*-test was used to compare the means between the study groups. The results were considered significant when the p value was <0.05.

Results

Of the 30 subjects injected with Mesigyna, 25 completed the study, whereas of the 30 subjects receiving TriNovum, 23 completed the study.

Effects on Glucose Metabolism

In four subjects of the Mesigyna group and in two of the TriNovum group, the variation coefficient of the euglycemic glucose clamp test, either at basal time or at the 6th cycle, was greater than 5. For this reason, only 22 women in each group were eligible for statistical analysis.

The two groups were tested for matching in age, height, weight, fasting glucose, and insulin levels, and rate of disappearance at admission. As indicated in Table 1, no significant difference was found for any of the five measured parameters.

The effects of treatment within the two groups are shown in Table 2. No significant difference within each group was observed in fasting glucose and insulin levels, and in glucose rate of disappearance. Body weight was the only parameter that significantly increased in the Mesigyna group ($p = 0.0046$).

Table 3 shows that no statistically significant dif-

Table 1. Homogeneity test at basal time of five major parameters between the group receiving the injectable contraceptive Mesigyna and that taking the oral preparation TriNovum

		Mesigyna	TriNovum	*t*-Test *t*	*t*-Test p
Age	*m*	26.14	26.09	0.047	0.96
	SD	3.17	3.19		
Height	*m*	161.64	161.27	0.20	0.84
	SD	3.26	7.78		
Weight	*m*	53.82	54.63	0.813	0.421
	SD	2.68	3.88		
Glucose	*m*	84.14	81.68	1.33	0.189
	SD	6.22	5.98		
Insulin	*m*	10.14	10.10	0.16	0.87
	SD	2.87	2.79		
Rd	*m*	7.27	7.23	0.228	0.820
	SD	0.64	0.69		

Means (*m*) ± standard deviations (SD) for age (years), height (cm), weight (kg), fasting glucose (mg/dl), and insulin (μU/ml) levels, and rate of disappearance (Rd, mg/kg/min). *t* = Student's test; p = statistical significance.

ference was found when comparing the two groups in terms of effect of treatment.

Effects on Lipid Metabolism

The two groups were matched for cholesterol, HDL-cholesterol (HDL-C), LDL-cholesterol (LDL-C), VLDL-cholesterol (VLDL-C), and triglycerides at basal time. As indicated in Table 4, no significant difference was found between the two groups for HDL-C, VLDL-C, and triglycerides, while a significant difference was found for cholesterol (p = 0.0005) and LDL-C (p = 0.0018) levels.

Table 5 summarizes the mean values and standard deviations for cholesterol, HDL-C, LDL-C, VLDL-C, and triglycerides in subjects treated with Mesigyna and TriNovum at time 0 and after 6 treatment cycles.

A significant increase in total cholesterol, HDL-C, and VLDL-C was observed in both groups. No significant change in LDL-C and triglycerides occurred in the Mesigyna group, whereas TriNovum significantly increased plasma levels of triglycerides, while producing no significant change in LDL-C.

When comparing the two formulations in terms of effects on lipid metabolism, TriNovum induced a more significant increase with regard to cholesterol (p = 0.001), VLDL-C (p = 0.01), and triglycerides (p = 0.02), as shown in Table 6. In view of the lack of homogeneity between the two groups with respect of cholesterol and LDL-cholesterol (see Table 4), these differences are open to question.

Table 2. Effects of the two treatments on carbohydrate metabolism

		Mesigyna *T0*	Mesigyna *T6*	TriNovum *T0*	TriNovum *T6*
Weight	*m*	53.82	54.32	54.63	54.86
	SD	2.68	2.64	3.88	3.78
	t		3.169		1.742
	p		0.0046		0.09
Glucose	*m*	84.14	82.64	81.68	82.04
	SD	6.22	6.63	5.98	3.33
	t		1.086		0.35
	p		0.29		0.73
Insulin	*m*	10.14	10.14	10.0	9.86
	SD	2.87	2.25	2.79	1.70
	t		0.00		0.263
	p		1.00		0.795
Rd	*m*	7.27	7.40	7.23	7.29
	SD	0.64	0.50	0.69	0.57
	t		1.773		1.19
	p		0.09		0.247

Results are expressed as means (*m*) ± standard deviations (SD) at time 0 (*T0*) and after 6 treatment cycles (*T6*) for weight (kg), fasting glucose (mg/dl), insulin (μU/ml), and rate of disappearance (Rd, mg/kg/min). *t* = Student's test; p = statistical significance.

Table 3. Comparison of the two groups in terms of effects of treatment on carbohydrate metabolism

		Mesigyna vs TriNovum
Glucose	*t*	1.03
	p	0.31
Insulin	*t*	0.58
	p	0.56
Rd	*t*	0.17
	p	0.97
Weight	*t*	0.66
	p	0.51

t = Student's test; p = statistical significance.

Discussion

Metabolic effects of the long-acting progestin norethisterone enantate (NET-EN) have been extensively investigated and recently reviewed.[17,18] According to the majority of the data available, NET-EN does not significantly alter carbohydrate metabolism,[19–22] although in one study, an increase in blood glucose levels was found after 6 months of use, which reverted to normal after 12 months,[19] and another study showed a persistent and significant elevation of serum insulin levels.[20] There is, however, one study in which a decrease of around 20% in plasma HDL-cholesterol was reported following the injection of NET-EN.[23,24]

Metabolic changes induced by orally administered estrogen–progestin preparations containing norethisterone (NET) have also been reported. Early studies with high-dose oral preparations showed an increase in insulin but not in glucose levels.[25] More recent

Contraception
1997;55:295–300

Table 4. Homogeneity test for lipid parameters between the two groups at basal time

		Mesigyna	TriNovum	*t*-Test	
				t	p
Cholesterol	*m*	181.03	170.63	3.68	0.0005
	SD	10.71	11.14		
HDL-C	*m*	47.77	47.67	0.081	0.93
	SD	4.34	5.22		
Triglycerides	*m*	83.03	83.90	0.34	0.73
	SD	9.64	19.80		
VLDL-C	*m*	16.57	16.87	0.59	0.55
	SD	1.91	1.97		
LDL-C	*m*	116.70	106.10	3.27	0.0018
	SD	11.73	13.26		

Means (*m*) ± standard deviations (SD) for cholesterol, HDL-cholesterol, LDL-cholesterol, VLDL-cholesterol, and triglycerides (all expressed as mg/dl). *t* = Student's test; p = statistical significance.

data on triphasic formulations show a slight increase in glucose at 3–6 months and a slight increase in mean insulin at 3 months, although these increases are much smaller in magnitude than those previously observed with high-dose preparations.[26] Finally, in a randomized double-blind, WHO-sponsored study conducted with two low-dose combined oral contraceptives containing either NET or levonorgestrel, Loke et al.[27] found that NET-containing preparations induced a decrease in fasting and 2-h glucose levels.

Burkman, in 1989,[28] evaluated three different combinations of NET and ethinylestradiol and found, considering all preparations, an increase in total cholesterol, triglycerides, and LDL, with a decrease in HDL; interestingly enough, women on the lowest dose preparation had the smallest trend toward outliner status for HDL-C. In 1993, Kakis et al.[29] compared two triphasic pills, one containing dl-norgestrel, the other NET. The NET-containing triphasic caused an elevation in total triglycerides and in the LDL fraction; it also caused an increase in total cholesterol and in the LDL fraction. In contrast to this, Notelovitz et al.[30] did not find any significant change from baseline in total cholesterol, HDL-C, or LDL-C ratios.

Only preliminary investigations have been carried out to evaluate the metabolic impact of the monthly injectable Mesigyna. Haiba et al.[31] studied some metabolic parameters before and after 6 months of use and found no significant changes in the 2-h postprandial glucose levels, although no glucose tolerance test or insulin measurements were made. Jiang et al.[32] studied carbohydrate metabolism in 33 Chinese women; plasma glucose and the area under the oral glucose tolerance curve were slightly increased at 9

Table 5. Effects of treatment on lipid metabolism

		Mesigyna		TriNovum	
		T0	*T6*	*T0*	*T6*
Cholesterol	*m*	181.03	185.80	170.63	182.96
	SD	10.71	8.81	11.14	11.55
	t		4.285		7.982
	p		0.0003		0.000
HDL-C	*m*	47.77	51.76	47.66	53.30
	SD	4.34	5.23	5.22	5.81
	t		4.82		5.20
	p		0.0001		0.000
Triglycerides	*m*	83.03	87.32	83.90	98.69
	SD	9.63	19.65	19.80	11.64
	t		1.53		6.35
	p		0.138		0.000
VLDL-C	*m*	16.567	18.20	16.86	19.82
	SD	1.91	1.89	1.97	2.36
	t		7.22		6.15
	p		0.000		0.000
LDL-C	*m*	116.7	115.84	106.10	105.48
	SD	11.73	11.1	13.26	23.18
	t		0.401		0.27
	p		0.691		0.78

Results are expressed as means (*m*) ± standard deviations (SD) at time 0 (*T0*) and after 6 treatment cycles (*T6*) for total cholesterol, HDL-cholesterol, LDL-cholesterol, VLDL-cholesterol, and triglycerides (all expressed in mg/dl). *t* = Student's test; p = statistical significance.

Table 6. Comparison of the two groups in terms of effects of treatment on lipid metabolism

		Mesigyna vs TriNovum
Cholesterol	*t*	3.51
	p	0.001
HDL-C	*t*	0.93
	p	0.36
Triglycerides	*t*	2.46
	p	0.02
VLDC-C	*t*	2.59
	p	0.01
LDL-C	*t*	0.15
	p	0.88

t = Student's test; p = statistical significance.

months and returned to pretreatment levels within 3 months of discontinuation. Plasma insulin levels measured during the oral glucose tolerance test were significantly increased at 9 months and were still elevated 3 months after discontinuation, although all values were in the normal range.

In a prospective study, Kesserü et al.[33] compared serum lipid parameters in controls, women on a triphasic pill containing gestodene, and women injected with Mesigyna. They found that HDL-C did not change significantly between the groups, while LDL-C decreased progressively and significantly up to 12 months in users of the monthly injectable, with an increased HDL/LDL ratio. In contrast to this, a study conducted in China by Yang et al.[34] in women using Mesigyna over a period of 12 months, showed that HDL-C was significantly reduced while LDL-C remained unchanged, with a resultant reduction in the HDL/LDL ratio. Finally, in a large, multicenter, WHO-sponsored study,[35] Mesigyna and Cyclofem (depo-medroxyprogesterone acetate 25 mg + estradiol cypionate 5 mg) were compared for the effects on lipid metabolism. Both injectables reduced the HDL/LDL ratio, while Mesigyna significantly reduced triglycerides.

The present investigation confirms that the two preparations containing NET do not cause any significant deterioration in glucose metabolism. Indeed, no significant variations occurred not only in the fasting glucose and insulin levels but also in the euglycemic glucose clamp test, one of the most comprehensive methods available today for the evaluation of glucose metabolism. No statistically significant difference was found between the injectable and the oral formulation in terms of effects on carbohydrate metabolism; in addition, the significant increase in body weight observed in the Mesigyna group appears to be negligible from the clinical point of view, although it has been documented that, after age 18, weight increases, even within the normal range, are associated with increased cardiovascular disease.[36] All will depend on whether, after 6 months, the trend continues or is reversed.

With regard to the effect that the two preparations had on lipid metabolism, both EP combinations significantly increased total cholesterol, HDL-C, and VLDL-C, whereas no significant change was observed for LDL-C. Triglycerides were significantly increased only in the TriNovum group. Although it is not known whether these differences could be due to the lack of homogeneity between the two groups with respect of cholesterol and LDL-cholesterol, a likely explanation for the observed rise in triglycerides in the oral preparation only is the first pass effect.[34,35] It must be stressed, however, that since the tests at 6 months were carried out during the last week of the last cycle of administration, for both preparations, the observed changes occurred at a time when, presumably, the positive estrogenic effect was the weakest. At any rate, these changes are not likely to have any significant clinical consequence.

Acknowledgment

This research was supported by a grant from Schering AG (Berlin, Germany).

References

1. Pincus G, Garcia CR, Rock J, et al. Effectiveness of an oral contraceptive. Science 1959;130:81.
2. Siegel I. Conception control by long-acting progestogens: preliminary report. Obstet Gynecol 1963;21:666.
3. World Health Organization, Special Programme of Research, Development and Research Training in Human Reproduction. Tenth Annual Report, 1981.
4. Oriowo MA, Landgren B-M, Stenstrom B, Diczfalusy E. A comparison of the pharmacokinetic properties of three estradiol esters. Contraception 1980;21:415.
5. World Health Organization, Task Force on Long-Acting Systemic Agents for Fertility Regulation, Special Programme of Research, Development and Research Training in Human Reproduction. A multicentred phase III comparative study of two hormonal contraceptive preparations given once-a-month by intramuscular injection: I. Contraceptive efficacy and side effects. Contraception 1988;37:1.
6. Sang GW, Shao QX, Ge RS, et al. A multicentred phase III comparative clinical trial of Mesigyna, Cyclofem and Injectable No. 1 given by intramuscular injection to Chinese women. I. Contraceptive efficacy and side effects. Contraception 1995;51:167.
7. Sang GW, Shao QX, Ge RS, et al. A multicentred phase III comparative clinical trial of Mesigyna, Cyclofem and injectable no. 1 given by intramuscular injection to Chinese women. II. The comparison of bleeding patterns. Contraception 1995;51:185.
8. Gaspard UJ, Lefebvre PJ. Clinical aspects of the relationship between oral contraceptives, abnormalities of carbohydrate metabolism and the development of cardiovascular disease. Am J Obstet Gynecol 1990;163:334.
9. Ducimetière P, Eschwage E, Papoz L, Richard JL, Claude JR, Roschin GE. Relationship of plasma insulin levels to the incidence of myocardial infarction and coronary heart disease mortality in a middle-aged population. Diabetologia 1990;19:205.
10. La Rosa JC. The varying effects of progestins on lipid levels and cardiovascular disease. Am J Obstet Gynecol 1988;158:1621.
11. Newman WP, Freedman DS, Voors AW, et al. Relation of serum lipoprotein levels and systolic blood pressure to early atherosclerosis. The Bagalusa Heart Study. N Engl J Med 1986;314:138.
12. Hirvonen E, Malkonen M, Manninen V. Effects of different progestogens on lipoproteins during post menopausal replacement therapy. N Engl J Med 1981;304:560.
13. De Fronzo R, Tobin J, Andres R. Glucose clamp tech-

nique: a method for quantifying insulin secretion and resistance. Am J Physiol 1979;237:E214.
14. Henry RJ, Cannon DS, Winckelman JW. Clinical Chemistry. Principles and Techniques. Hagerstown: Harper & Row, 1974.
15. Cooper C. Manual of Laboratory Operations. Lipid Research Clinic Program. Vol. I. National Heart and Lung Institute, USA, 1974.
16. Stambury JB, Wyngaarden JB, Fredrickson DS. The Metabolic Basis of Inherited Disease. 2nd Ed. New York: McGraw-Hill Book Co., 1966.
17. Fahmy K, Abdel-Razik M, Shaaraway M, et al. Effect of long-acting progestagen-only injectable contraceptives on carbohydrate metabolism and its hormonal profile. Contraception 1991;44:419.
18. McEwan JA, Griffin M, Fotherby K, Trayner I. Long-term use of depot-norethisterone enanthate: effect on blood lipid fractions. Contraception 1992;46:49.
19. Dhall K, Kumar M, Rastogi G, Devi P. Short-term effects of norethisterone enanthate and medroxyprogesterone acetate on glucose, insulin, growth hormone and lipids. Fertil Steril 1977;28:156.
20. Amatayakul K, Suriyanon V. The effect of long-acting injectable contraceptives on carbohydrate metabolism. Int J Gynecol Obstet 1985;23:361.
21. Howard G, Elder MG. The effect of intramuscular norethisterone enanthate on intravenous glucose tolerance and blood coagulation factors VII and V. Br J Obstet Gynaecol 1977;84:618.
22. Howard G, Blair M, Fotherby K, Trayner I, Hamawi A, Elder MG. Some metabolic effects of long-term use of the injectable contraceptive norethisterone enanthate. Lancet 1982;1:423.
23. Fotherby K, Trayner I, Howard G, Hamawi A, Elder MG. Effect of injectable norethisterone oenanthate (Norigest) on blood lipid levels. Contraception 1982;25: 435.
24. Fotherby K. Metabolic effects of injectable contraceptives. Br J Clin Pract 1982;17(Suppl.):2G.
25. Spellacy WN. Carbohydrate metabolism in male infertility and female fertility-control patients. Fertil Steril 1976;27:1132.
26. Bowes WA, Katta LR, Droegemueller W, Bright TG. Triphasic randomized clinical trial: comparison of effects on carbohydrate metabolism. Am J Obstet Gynecol 1989;161:1402.
27. Loke DFM, Ng CSA, Holck S, Hall PE, Ratnam SS. Lipid and biochemical changes after low-dose oral contraception. Contraception 1992;46:227.
28. Burkman RT, Zacur HA, Kimball AW, Kwiterovich P, Bell WR. Oral contraceptives and lipids and lipoproteins. Part II: relationship to plasma steroid levels and outlier status. Contraception 1989;40:675.
29. Kakis G, Powell M, Marshall A, Steiner G. A randomized, comparative open study of the effects of two oral contraceptives, Triphasil and Ortho 7/7/7, on lipid metabolism. Contraception 1993;47:131.
30. Notelovitz M, Levenson I, McKenzie L, Khan FY. The effect of low-dose oral contraceptives on lipid and lipoproteins in two at-risk populations: young female smokers and older premenopausal women. Contraception 1991;44:505.
31. Haiba NA, El-Habashy MA, Said SA, Darwish EA, Abel-Sayed WS, Nayel SE. Clinical evaluation of two monthly injectable contraceptives and their effects on some metabolic parameters. Contraception 1989;39: 619.
32. Jiang HY, Yang H, Zhang MY, Lu FY, Gu LJ. Effects of long-term use of two monthly injectable contraceptives on carbohydrate metabolism and liver function. Chin J Fam Plann 1992;1:28.
33. Kesserü E, Albornoz H, Diaz M, Socolsky R. Serum lipid variations in women on copper intrauterine devices, triphasic oral contraceptives and monthly injectable contraceptives. Contraception 1991;44:13.
34. Yang H, Zhang MY, Lu FY, Jiang HY. The effect of a monthly injectable contraceptive on serum lipids in Chinese women. Reprod Contraception (Beijing) 1994; 14:427.
35. WHO Special Programme of Research, Development and Research Training in Human Reproduction. Comparative study of two once-a-month injectable steroidal contraceptives (Mesigyna and Cyclofem) on lipid, lipoprotein, and apolipoprotein metabolism. Contraception. In press.
36. Manson JE, Willett WC, Stampfer MJ, et al. Body weight and mortality among women. NE J Med 1995; 333:677.